# Real-World Coordinate-Plane Problems; Perimeter

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lesson 6 — Real-World Coordinate-Plane Problems; Perimeter | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.2 |

## 2. Standards

**MA.8.GR.1.2** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

**Must.** Find the perimeter of a figure on the coordinate plane, each slanted side by its right triangle (a triangle with no right angle split by its height); the straight-line and along-the-grid distances on a map, and how much shorter the straight line is; an irrational total kept exact and rounded once.

**Must not.** The distance formula (Clarification 2); radicands over 225 or with a square factor; combining radicals (√34 + √34 is two lengths added, not 2√34 — Algebra 1).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes I and III: one perimeter as sides on the grid, as right triangles and as a sum |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.7.1 | Apply mathematics to real-world contexts. | Notes II and boards 5, 7 and 8: maps, the streets against the straight line |

## 4. Learning Target and Essential Question

**Learning target.** I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

**Essential question.** How do you measure around a shape whose sides do not follow the grid lines?

**Building on.** 6.05 — the distance between two points on the grid; 6.03 — the Pythagorean Theorem; Grade 6 (MA.6.GR.1.3) — the perimeter of a rectangle on the coordinate plane.

**Working toward.** 6.07–08 — similar triangles; Geometry — perimeters on the plane, and the distance formula, which is this triangle written as one line.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front. Expand once per law; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why is the straight path shorter than the streets unless the two points share an x or a y? | DOK 2 |
| Notes III rounded once, at the end. When does rounding each side first change the answer? | DOK 3 |
| A triangle has no right angle. Where would you draw a line to give it two? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$. Find its perimeter. | ***30 units*** | 17 — left the slanted side out [Notes I: every side]; 34 — took AC as 12 + 5 [MA.8.GR.1.2: the theorem, not a sum] |
| 2 | A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$. Find its perimeter, to the nearest tenth. | ***22.8 units*** | 13 — left the slanted side out [Notes I]; 22.9 — rounded 9.849 to 9.85 and then to 9.9 [one rounding, at the end — Notes III]; 26 — the slanted side as 4 + 9 [MA.8.GR.1.2] |
| 3 | Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle. Find its perimeter. | ***42 units*** | 40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]; a perimeter with 12 in it — the height is inside the triangle, not a side [B1G-M, GR.1.2: the theorem on a triangle with no right angle] |
| 4 | What is the perimeter of the triangle with corners (−2, 3), (2, 3) and (2, −4), to the nearest tenth? | ***D — 19.1 units*** | (A) 4 + 7: left the slanted side out [Notes I: every side]; (B) 4 + 7 + 11: the slanted side taken as the sum of the legs [MA.8.GR.1.2: the theorem, not a sum]; (C) 4 + 7 + 65: forgot the square root [B1G-M, GR.1.1: root errors] |
| 5 | On a map in blocks, the library is at $(-7, -1)$ and the pool at $(5, 4)$. How much shorter is the straight line than the streets? | ***4 blocks shorter*** | 13 or 17 — one path, not how much shorter [the question asks a difference]; 30 — added the two paths [the question asks a difference]; 2 across, from 7 − 5 [Source of Truth, GR.1.2: integer errors across an axis] |
| 6 | How long is a diagonal of the rectangle with corners (−3, −4), (6, −4), (6, 8) and (−3, 8)? | ***A — 15 units*** | (B) 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]; (C) 9 + 12 + 9 + 12: the perimeter, not one diagonal [Notes II: the straight line is one segment]; (D) 6 and 8, one corner's coordinates, taken as the legs [Math Nation names wrong horizontal and vertical lengths] |
| 7 | On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$. How far is it straight across, to the nearest tenth? | ***$\sqrt{106} \approx 10.3$ m*** | 14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]; √26 — 4 − 5 = 1 for the across leg [Source of Truth, GR.1.2: integer errors across an axis] |
| 8 | On a trail map in miles, the camp is at $(-3, -6)$, the lake at $(2, 6)$ and the trailhead at $(-1, 0)$. How far is the camp from the lake, in a straight line? | ***13 mi*** | 17 — added the legs [MA.8.GR.1.2]; a distance to the trailhead — the wrong pair of points |

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 9 | Mia says the segment from (0, 0) to (3, 1) is 3 units long, because it crosses 3 grid squares. Is she right? Explain, and find its length to the nearest tenth. | ***No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.*** | 'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]; 4 — added the legs [MA.8.GR.1.2] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** A printed grid for every board. Find the sides on grid lines first and write them on the figure; then the slanted side, by its triangle. "Every side counts" said aloud with every perimeter.

**ELL.** **Perimeter**: around the outside — every side. **Along the streets**: on the grid lines. **Straight line**: one segment, as a bird flies. Trace each with a finger on the grid.

**Enrichment.** Plot (0, 0), (6, 0) and (3, 4). Is it a right triangle? (No: 25 + 25 is not 36.) Find its perimeter. (5 + 5 + 6 = 16.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: Mia counted the squares her segment crosses, but a slanted segment is the hypotenuse of a right triangle — legs 3 and 1, √10 ≈ 3.2, longer than 3. A board that says only '3.2' has the length and not the reason.

**Vocabulary.** **perimeter**; **height of a triangle**; **straight-line distance**

**Materials.** Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Pythagorean theorem: find the perimeter — VGE.